



SYMBIOSIS INTERNATIONAL (DEEMED UNIVERSITY)

(Established under section 3 of the UGC Act, 1956)

Re-accredited by NAAC with 'A++' Grade | Awarded Category - I by UGC

Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

Course Name: Introduction to Image Processing
Course Code: TE7428
Faculty: Engineering
Course Credit: 3
Course Level: 3
Sub-Committee (Specialization): Electronics and Telecommunication
Learning Objectives:

The student will be able to:

Demonstrate understanding of basic concepts of two dimensional signal acquisition, sampling and quantization.

Demonstrate understanding of fundamental image manipulations such as histogram stretching, histogram equalization, contrast enhancement, log transforms and image negatives.

Demonstrate understanding of spatial filtering techniques.

Demonstrate understanding of 2D Fourier transform, 2D FFT and filtering images in Fourier domain.

Demonstrate understanding of image morphology that includes image manipulation using set theory.

Understand concepts in image segmentation.

Describe the basic issues and the scope of image processing, and the roles of image processing and systems in a variety of applications.

Select an appropriate approach to a given image processing task and critically evaluate and benchmark the performance of possible alternative techniques.

Books Recommended:

Book	Author	Publisher
Digital Image Processing	Gonzalez, Woods	PHI, 2nd edition
Digital Image Processing	Pratt W.K.	John Wiley, 2001
Fundamentals of Digital Image Processing	Jain A.K.	PHI, 1997
Image Processing, Analysis & Machine Vision	Milan Sonka	Thomson Publication

Course Outline:

Sr. No.	Topic	Actual Teaching Hours	Contact Hours Equivalence
1	Digital Image Fundamentals Digital Image representation Elements of Digital Image Processing Systems. Elements Of Visual perception structure of human eye simple image model sampling and quantization basic relationship between pixels imaging geometry photographic film.	6	6
2	Contrast enhancement and spatial filtering and image restoration Histogram Histogram Equalization	10	10

	Spatial filtering: averaging weighted averaging Gaussian filter , Laplacian, Sobel, Prewitt, Order Statistic filters. Noise models noise removal filters- harmonic contra-harmonic order statistic adaptive median filter. Degradation function modeling weiner filter inverse filtering.		
3	Filtering in Fourier Domain 2-D Discrete Fourier Transform FT properties Filtering in FD: Low pass filtering high pass filtering band pass and band reject filters notch filters homomorphic filters Gaussian butterworth filters.	8	8
4	Segmentation, point, line and edge detection Thresholding Otsu's method Global thresholds region splitting and merging Hough Transform Watersheds.	7	7
5	Morphological Image Processing Neighborhood concepts adjacency and distance measures dilation & erosion opening & closing operations basic morphological operations such as region filling thinning thickening skeletons pruning for binary and gray scale images	8	8
6	Image Processing Applications Applications of image processing in biometrics such as face recognition figure print recognition Medical applications such as tumor detection Magnetic Resonance Imaging analysis using transforms Morphological applications.	6	6
Total		45	45

Pre Requisites:

Signals and Systems
 Digital Signal Processing
 MATLAB/Python basics

Evaluation:

- A. Continuous Assessment
 - a. Unit Tests
 - b. Class Tests
 - c. Mini Project report
 - d. Quizzes
 - e. Seminars

B. End Semester Examination

Pedagogy:

Lectures

Problem solving using MATLAB

Mini projects